

AI Security Investigation Report

Multiple SSH authentication failures detected against privileged account root from external source.

Platform: wazuh

Evidence-grounded security investigation summary

Investigation Status

DISPOSITION	SEVERITY	CONFIDENCE	ANALYST REVIEW	MITRE
Suspicious Activity	high	82	Required	T1110

Alert Overview

Platform	wazuh
Alert Type	ssh_failed_login
Vendor Severity	high
Confidence	82
Source IP	203.0.113.45
Destination IP	10.0.1.15
Hostname	prod-web-01
Username	root

Executive Summary

wazuh reported a ssh_failed_login alert (high severity). Multiple SSH authentication failures detected against privileged account root from external source. Key observables: source IP 203.0.113.45, destination IP 10.0.1.15, host prod-web-01, user root.

Recommended Disposition

Suspicious Activity

The normalized alert contains authentication-failure activity and identifying network, account, or host context. Successful access or downstream compromise cannot be confirmed from the available evidence.

Supporting Evidence: EVID-002, EVID-004, EVID-007, EVID-005, EVID-006, EVID-008

Analyst Review Required: Yes

Evidence

ID	Kind	Category	Evidence	Value	Source	Context
EVID-001	metadata	Alert Metadata	Platform	wazuh	alert.platform	Platform that generated the normalized alert.
EVID-002	metadata	Alert Metadata	Alert type	ssh_failed_login	alert.alert_type	Normalized alert type.
EVID-003	metadata	Alert Metadata	Vendor severity	high	alert.severity	Severity reported by the alert source.
EVID-004	metadata	Alert Metadata	Description	Multiple SSH authentication failures detected against privileged account root from external source.	alert.description	Description supplied with the normalized alert.
EVID-005	observable	Identity	Username	root	alert.observables.username	Account associated with the alert.
EVID-006	observable	Host	Hostname	prod-web-01	alert.observables.hostname	Host associated with the alert.
EVID-007	observable	Network	Source IP	203.0.113.45	alert.observables.source_ip	Source IP associated with the alert.
EVID-008	observable	Network	Destination IP	10.0.1.15	alert.observables.destination_ip	Destination IP associated with the alert.

Analyst Reasoning

Observations

OBS-001

The normalized alert reports SSH authentication failures.

Evidence: EVID-002, EVID-004

OBS-002

The alert contains a source IP.

Evidence: EVID-007

OBS-003

The alert contains a destination IP.

Evidence: EVID-008

OBS-004

The alert contains a username.

Evidence: EVID-005

OBS-005

The alert contains a hostname.

Evidence: EVID-006

Assessment

ASM-001

The reported behavior is consistent with repeated SSH authentication failures.

Evidence: EVID-002, EVID-004

ASM-002

The normalized evidence does not establish that authentication succeeded.

Alternative Explanations

ALT-001

Automated internet scanning or password guessing.

ALT-002

Authorized security testing or administrative validation.

ALT-003

A vendor detection triggered on repeated but unsuccessful authentication activity.

Evidence Gaps

GAP-001

No successful-authentication telemetry is included in the normalized alert.

GAP-002

No post-authentication process telemetry is included.

GAP-003

No lateral-movement telemetry is included.

Confidence Rationale

Supporting Factors

SUP-001

Authentication-failure activity is reported in the normalized alert.

Evidence: EVID-002, EVID-004

SUP-002

A source IP is identified in the normalized evidence.

Evidence: EVID-007

SUP-003

A target username is identified.

Evidence: EVID-005

SUP-004

A destination host is identified.

Evidence: EVID-006

SUP-005

A destination IP is identified.

Evidence: EVID-008

Limiting Factors

LIM-001

Successful authentication cannot be confirmed from normalized evidence alone.

LIM-002

Post-authentication endpoint activity is not available in the normalized alert.

LIM-003

Containment or response status is not available.

Overall

The available normalized identifiers provide grounding for the authentication investigation, while successful-login and post-authentication telemetry remain unavailable. These factors provide context for the reported confidence score but do not reproduce its calculation.

Severity Assessment

Vendor severity 'high' indicates elevated priority. Failed SSH logins warrant review for brute-force activity, but impact depends on whether authentication eventually succeeded and which accounts were targeted.

MITRE ATT&CK Mapping

Technique ID	Technique name	Tactic	Confidence	Rationale
T1110	Brute Force	Credential Access	medium	Repeated or failed SSH authentication attempts align with credential brute-force behavior; confirm volume, targeting, and success before escalation.

Recommended Investigation Queries

QRadar AQL

```
SELECT sourceip, destinationip, username, LOGSOURCETYPENAME(devicetype), starttime, category, qid FROM events WHERE sourceip = '203.0.113.45' LAST 24 HOURS
```

```
SELECT sourceip, destinationip, destinationport, starttime FROM events WHERE destinationip = '10.0.1.15' LAST 24 HOURS
```

```
SELECT LOGSOURCETYPENAME(devicetype), sourceip, username, starttime, category FROM events WHERE devicename ILIKE '%prod-web-01%' LAST 7 DAYS
```

```
SELECT username, sourceip, LOGSOURCETYPENAME(devicetype), starttime, category FROM events WHERE username ILIKE '%root%' LAST 7 DAYS
```

```
SELECT sourceip, username, COUNT(*) AS failures FROM events WHERE category = 1515 AND LOGSOURCETYPENAME(devicetype) ILIKE '%Linux%' LAST 24 HOURS GROUP BY sourceip, username HAVING COUNT(*) > 10
```

Microsoft Sentinel KQL

```
SecurityEvent
| where TimeGenerated > ago(24h)
| where IPAddress == "203.0.113.45" or SourceIP == "203.0.113.45"
| project TimeGenerated, Computer, Account, IPAddress, Activity
```

```
CommonSecurityLog
| where TimeGenerated > ago(24h)
| where DestinationIP == "10.0.1.15"
| summarize count() by DeviceVendor, DeviceProduct, Activity
```

```
Heartbeat
| where TimeGenerated > ago(7d)
| where Computer contains "prod-web-01"
| summarize arg_max(TimeGenerated, *) by Computer
```

```
SigninLogs
| where TimeGenerated > ago(7d)
| where UserPrincipalName contains "root" or Account contains "root"
| project TimeGenerated, UserPrincipalName, IPAddress, ResultType, AppDisplayName
```



```
SecurityEvent
| where TimeGenerated > ago(24h)
| where EventID == 4625
| summarize FailureCount = count() by IPAddress, Account
| where FailureCount > 10
```

Microsoft Defender Advanced Hunting KQL

```
DeviceNetworkEvents
| where Timestamp > ago(24h)
| where RemoteIP == "203.0.113.45" or InitiatingProcessRemoteIP == "203.0.113.45"
| project Timestamp, DeviceName, ActionType, RemoteIP, InitiatingProcessFileName
```

```
DeviceNetworkEvents
| where Timestamp > ago(24h)
| where RemoteIP == "10.0.1.15"
| project Timestamp, DeviceName, RemoteIP, RemotePort, InitiatingProcessFileName
```

```
DeviceInfo
| where DeviceName contains "prod-web-01"
| project Timestamp, DeviceName, OSPlatform, PublicIP, ExposureLevel
```

```
DeviceLogonEvents
| where Timestamp > ago(7d)
| where AccountName contains "root"
| project Timestamp, DeviceName, AccountName, LogonType, RemoteIP
```

```
DeviceLogonEvents
| where Timestamp > ago(24h)
| where LogonType in ('RemoteInteractive', 'Network')
| where ActionType == 'LogonFailed'
| summarize FailureCount = count() by RemoteIP, AccountName
```

OpenSearch / DQL

```
source.ip: "203.0.113.45" AND @timestamp:[now-24h TO now]
```

```
destination.ip: "10.0.1.15" AND @timestamp:[now-24h TO now]
```

```
host.name: "prod-web-01" AND @timestamp:[now-7d TO now]
```

```
user.name: "root" AND @timestamp:[now-7d TO now]
```

```
event.action: "authentication_failure" AND event.category: "authentication" AND @timestamp:[now-24h TO now]
```

Next Investigation Steps

- Validate whether any successful SSH logins occurred from the source IP after failures.
- Check if the targeted username is a valid local account or a common brute-force target (root, admin).
- Review firewall and geo-IP context; block or rate-limit if attack volume is sustained.
- Search for lateral movement or credential reuse if a successful login is confirmed.

Detection Engineering Opportunities

- Threshold-based detection for repeated SSH authentication failures from a single source IP.
- Correlation rule linking failed logins to successful logins from the same source within a short window.
- Geo-velocity or impossible-travel check if successful authentication follows brute-force patterns.

Analysis Limitations

- Offline v1 framework — no live SIEM, EDR, or email security API queries were executed.
- MITRE mappings are deterministic templates, not ML-classified or vendor-validated attributions.
- Recommended queries are example pivots; field names and log sources vary by deployment.

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This report supports analyst review and does not perform automated incident closure or containment.